

Beverly Xu

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EDUCATION

Harvey Mudd College | B.S. in Engineering

Relevant coursework

GPA: 3.9

Digital Elec & Comp Engineering · Intro to Engr Design/Manufacturing · Electromagnetic Theory & Optics

PUBLICATIONS

"cHyRRT and cHySST: Two Motion Planning Tools for Hybrid Dynamical Systems," IEEE CASE, August 2025

– First author and presenter, implemented both planners with acceptance into Open Motion Planning Library.

ROBOTICS EXPERIENCE

Electrical/Mechanical Engineer, Harvey Mudd RoboSub (AUV) Team, Claremont, CA September 2025 - Present

- Designed three-stage 25-40 kHz hydrophone signal amplifier with 20,000-120,000x adjustable gain, DC bias stages to preserve signal through single-rail op-amps and protect Teensy input, and low-pass filter removing noise above the expected frequencies.
- Documented and integrated amplifier subcircuit into main signal board. Designed in KiCAD and simulated using LTSpice.
- Designed torpedo launcher using SolidWorks FEA and flow simulation to model force distribution and fluid dynamics. Manufactured 2-inch diameter launcher within 2% tolerance using Fusion CAM and CNC bed mill.

Software Lead, Mechanical Science and Engineering Robotics Team, Saratoga, CA August 2021 - June 2025

- Led group of 6 in control systems implementation (PID, odometry, etc.), sensor fusion, and full-stack robot troubleshooting.
- Facilitated adoption of LiveShare to reduce onboarding time and merge conflicts, while diversifying code contributors.

RESEARCH EXPERIENCE

Intern, Lab for Investigations of Local Air Quality, Claremont, CA September 2025 - Present

- Designed and prototyped CO2 level indicator with 80% cost reduction from comparable commercial option, and modeled electronics box in SolidWorks using edit-in-context assembly workflow.
- Developed Linux, BASH, and IGOR scripts for Pi and Windows devices to stream data to Firebase from 35 PurpleAir and meteorological sensors running legacy middleware.

NSF-funded Researcher, RoboLand Lab @ USC, Los Angeles, CA May 2026 - August 2026

- Developed locomotion testbed with spring-loaded obstacles, mounted on custom sliders. Enabled repeatability and durability for 100+ trials. Improved the design of caterpillar-inspired robot to decrease gear slack by 50% and simplify inverse kinematics.
- Formulated research question on legged locomotion, leveraging spring-loaded obstacles. Tested hypothesis across 100+ trials using MoCap motion tracking system, and cleaned and analyzed data points in MATLAB and Python.

Intern, UC Santa Cruz, Hybrid Systems Lab, Santa Cruz, CA August 2023 - August 2025

- Developed 2 motion planners for hybrid dynamical systems (collision-resilient quadcopter, bouncing ball). Benchmarked for planning time, success rate, and node count, achieving 84% reduction in planning time and $5.4\times$ success rate versus baseline.
- Authored C++ tests to validate motion plans in ROS 2 and VM image for conference submission.

PROJECTS

Record player with motorized turntable, Claremont, CA July 2026 - Present

- Designed and simulated power electronics and control systems board for record sensing, audio, lighting, battery charging, and motor control. Used KiCAD for schematic capture and PCB layout.
- Planning to solder SMD components onto bare board using stencil and reflow.

Analog alarm clock with single motor-actuated hands, Claremont, CA April 2026 - May 2026

- Designed custom gear box in SolidWorks with 1:60:360 gear ratios, tested dynamics using motion study.

Dance floor tile electricity generator, Saratoga, CA January 2025 - May 2025

- Designed mechanism with 150 lbs max load to backdrive motor, leveraging cross-functional collaboration with a team of 3.

TECHNICAL SKILLS

Mechanical/Manufacturing: Advanced in SolidWorks and 3D printing, Intermediate in Fusion, 3D, CAM, DFM

Electrical: Advanced in KiCAD, Intermediate in embedded systems, mixed-signal design, signal processing, circuit simulation, LTSpice, PCB layout, power electronics, control systems

Software: Advanced in C++, closed-loop control, Intermediate in Python, LaTeX, Java, Linux, Git, motion planning

HONORS

Society of Women Engineers Motorola Foundation Scholarship 2025

Alltricity Foundation Scholarship 2025

Harvey Mudd Merit Scholarship 2024 - Present